DO NOW — RECALL

byVal, byRef and modular programming

Name: _____ Class: _____ Date: _____

1. What does return do in a C# function?

2. Can a method read a local variable declared inside another method?

3. What is a[0] for int[] a={9,6,3}?

4. If Double(n) returns 2*n, find Double(Double(3)).

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Finishes the invocation and returns a result to its caller.

2. Model answer

Not directly. It is outside the variable's scope.

3. Model answer

9.

4. Model answer

Double(3)=6; Double(6)=12.

Vocabulary — know these terms

by value — The subroutine receives a copy of the value.

by reference — The parameter refers to the caller's variable.

modularity — Divide a program into smaller testable subroutines.

ref — C# keyword enabling reference parameter passing.

side effect — A change to external program state.

NEW LEARNING

Read, annotate and explain

CALLER

The caller declares marks = 10.

ALIAS

ref makes score refer to the caller's marks rather than an independent copy.

AFTER

Changing score therefore changes marks, which becomes 15.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A function receives a copy of x=7 and adds 4. What is x afterwards?

Predict first

A new parameter value receives a copy of the integer 7.

Model the result

value becomes 11 inside AddFour, but x remains 7 after the call.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

A method takes int number by value and adds 2. Caller x=5. What is x after the call?

Question 2

Write a C# procedure that doubles an integer passed by reference.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain when returning a result is preferable to changing a ref parameter.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Changing a by-value int parameter always changes the caller's int.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.